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Semiconductor devices including voltage monitors

Abstract

The present disclosure provides a voltage monitor and a semiconductor device including the voltage monitor. The voltage monitor includes a first voltage-to-digital converter (VDC), a second VDC, a first digital-to-binary converter (DBC), a second DBC, and an adder. The first VDC is configured to generate a first digital signal in response to a clock signal, and the second VDC is configured to generate a second digital signal in response to the clock signal. The first DBC is connected to the first VDC, and configured to convert the first digital signal to a first binary signal. The second DBC is connected to the second VDC, and configured to convert the second digital signal to a second binary signal. The adder is connected to the first DBC and the second DBC, and configured to combine the first binary signal and the second binary signal into an output signal.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of prior-filed provisional application No. 63/384,954, filed on Nov. 24, 2022.

BACKGROUND

(1) The present disclosure relates to a semiconductor device including one or more voltage monitors to monitor power voltage (VDD).

(2) In semiconductor industries, VDD droop may result from chip activity, and can lead to timing violations and inhibit speed/performance. Generally, designs can define a high voltage margin to

minimize the effect of VDD droop on chip performance. However, the high voltage margin increases power consumption. Therefore, an improved accurate voltage monitor for monitoring VDD droop is required.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features can be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A shows an equivalent circuit of a semiconductor device, in accordance with some embodiments of the present disclosure.

(3) FIG. 1B is a graph illustrating time versus power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure.

(4) FIG. 2A shows an architecture of a voltage monitor for monitoring the power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure.

(5) FIG. 2B shows an architecture of a voltage monitor for monitoring the power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure.

(6) FIG. 3A is a flowchart showing operations of a voltage monitor, in accordance with some embodiments of the present disclosure.

(7) FIG. 3B is a flowchart showing operations of a voltage monitor, in accordance with some embodiments of the present disclosure.

(8) FIG. 3C is a flowchart showing operations of a voltage monitor, in accordance with some embodiments of the present disclosure.

(9) FIG. 4A shows an architecture of a voltage-to-digital converter included in a voltage monitor, in accordance with some embodiments of the present disclosure.

(10) FIG. 4B is a timing diagram of the voltage-to-digital converter depicted in FIG. 4A, in accordance with some embodiments of the present disclosure.

(11) FIG. 5A depicts an operating exemplary voltage-to-digital converter included in the voltage monitor, in accordance with some embodiments of the present disclosure.

(12) FIG. 5B depicts an operating exemplary voltage-to-digital converter included in the voltage monitor, in accordance with some embodiments of the present disclosure.

(13) FIG. 6 is a graph illustrating time versus power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure.

(14) FIG. 7 is a graph illustrating power voltage (VDD) versus binary signal of the recorded voltage of a semiconductor device, in accordance with some embodiments of the present disclosure.

(15) FIG. 8A shows an exemplary method for calibrating the voltage monitor of the semiconductor device, in accordance with some embodiments of the present disclosure.

(16) FIG. 8B shows an exemplary method for calibrating the voltage monitor of the semiconductor device, in accordance with some embodiments of the present disclosure.

(17) FIG. 8C shows an exemplary method for calibrating the voltage monitor of the semiconductor device, in accordance with some embodiments of the present disclosure.

(18) FIG. 9 is a top view of an exemplary semiconductor device, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(19) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components

and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features can be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(20) Further, it will be understood that when an element is referred to as being “connected to” or “coupled to” another element, it can be directly connected to or coupled to the other element, or intervening elements can be present.

(21) Embodiments, or examples, illustrated in the drawings are disclosed as follows using specific language. It will nevertheless be understood that the embodiments and examples are not intended to be limiting. Any alterations or modifications in the disclosed embodiments, and any further applications of the principles disclosed in this document are contemplated as would normally occur to one of ordinary skill in the pertinent art.

(22) Further, it is understood that several processing steps and/or features of a device can be only briefly described. Also, additional processing steps and/or features can be added, and certain of the following processing steps and/or features can be removed or changed while still implementing the claims. Thus, it is understood that the following descriptions represent examples only, and are not intended to suggest that one or more steps or features are required.

(23) In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(24) FIG. 1A shows an equivalent circuit **11** of a semiconductor device, in accordance with some embodiments of the present disclosure. The circuit **11** includes a voltage regulator **100** and a semiconductor device **110**. The semiconductor device **110** includes a first loop **111**, a second loop **112**, and a third loop **113**.

(25) In some embodiments, the semiconductor device **110** may include a substrate (such as printed circuit board (PCB)), and one or more semiconductor packages including one or more dies disposed on the substrate. The semiconductor device **110** can receive the power supply voltage (VDD) from the voltage regulator **100**. In some embodiments, the activity of the semiconductor device **110** causes power supply voltage (VDD) droop (also known as IR drop). The VDD droop can include three kinds of VDD droop, which will be discussed in accordance with FIG. 1B.

(26) FIG. 1B is a graph **12** illustrating time versus power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure. Referring to FIG. 1B, the x-axis represents time and y-axis represents voltage. The graph **12** shows line **121** and curve **122**. Line **121** shows the power supply voltage (also called the nominal voltage $V_{sub.nom}$) supplied to the semiconductor device **100**. In some embodiments, the nominal voltage $V_{sub.nom}$ can be a constant voltage. Therefore, the line **121** can be a horizontal line. The curve **122** shows the VDD of the semiconductor device **100** varying with time passage. The curve **122** can approach the line **121** at the beginning. In some embodiments, the curve **122** includes three droops (marked 1.sup.st Droop, 2.sup.nd Droop, and 3.sup.rd Droop).

(27) The first droop of the curve **122** may be determined by the resistance, capacitance, and inductance of the die and semiconductor package of the semiconductor device. In some embodiments, the first droop occurs due to the elements in the semiconductor device activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). The second droop of the curve **122** may be determined by the resistance, capacitance, and inductance of the semiconductor package and substrate (PCB) of the

semiconductor device. In some embodiments, the second droop occurs due to the elements in the semiconductor device activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). The second droop can result from the same reason of the first droop, but happen later than the first droop. The third droop of the curve **122** may be determined by the resistance, capacitance, and inductance of the semiconductor package and substrate (PCB) of the semiconductor device. In some embodiments, the third droop occurs due to the elements in the semiconductor device activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). The third droop can result from the same reason of the second droop, but happen later than the second droop.

(28) Referring back to FIG. **1A**, the voltage regulator **100** provides VDD to the semiconductor device **110**. The first loop **111** includes resistance, capacitance, and inductance of dies and semiconductor packages activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). In some embodiments, the first loop **111** can correspond to 1.sup.st Droop in FIG. **1B**. That is, 1.sup.st Droop occurs due to the first loop **111**. The second loop **112** includes resistance, capacitance, and inductance of semiconductor packages and substrates activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). In some embodiments, the second loop **112** can correspond to the 2.sup.nd Droop in FIG. **1B**. That is, the 2.sup.nd Droop occurs due to the second loop **112**. The third loop **113** includes resistance, capacitance, and inductance of semiconductor packages and substrates activated from a low current mode to a high current mode (for example, the semiconductor device is activated from idle). In some embodiments, the third loop **113** can correspond to the 3.sup.rd Droop in FIG. **1B**. That is, the 3.sup.rd Droop occurs due to the third loop **113**. FIG. **2A** shows an architecture of a voltage monitor **20** for monitoring the power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure. The voltage monitor **20** includes voltage-to-digital converters (VDC) **210-0**, **210-1**, **210-2** . . . and **210-n**, digital-to-binary converters (DBC) **220-0**, **220-1**, **220-2** . . . and **220-n**, an adder **230**, an overshoot detector **240**, an undershoot detector **250**, and a multiplexer (MUX) **260**.

(29) Referring to FIG. **2A**, the voltage monitor **20** can include two or more VDCs **210-0**, **210-1**, **210-2** . . . and **210-n**. For example, the voltage monitor **20** can include two, four, eight, or sixteen VDCs. The VDCs **210-0** and **210-n** can be configured to receive the clock signal CLK. In some embodiments, the VDCs **210-0** and **210-n** can be configured to generate a digital signal in response to a clock signal CLK. The VDC **210-0** is configured to generate a digital signal VDC_0 in response to the clock signal CLK. The VDC **210-n** is configured to generate a digital signal VDC_n in response to the clock signal CLK.

(30) In some embodiments, each VDC may include one or more delay cells to generate the digital signal in response to the clock signal CLK. The digital signals VDC_0 and VDC_n can be a 1-bit, 2-bit, 3-bit, 4-bit . . . or 16-bit signal. In some embodiments, the digital signal can be more than 16 bits. In some embodiments, the bit count of the digital signal can depend on the number of the delay cells (shown in FIG. **2B**) included in the VDCs **210-0** and **210-n**. In some embodiments, the form of the digital signal is not limited. For example, the digital signal can be in the form of thermometer code or one hot encoding.

(31) The voltage monitor **20** can include two or more DBCs **220-0**, **220-1**, **220-2** . . . and **220-n**. For example, the voltage monitor **20** can include two, four, eight, or sixteen DBCs. In some embodiments, the number of the DBCs can be identical to that of the VDCs. The DBCs **220-0** and **220-n** are connected to a corresponding VDC. For example, the DBC **220-0** can be connected to the VDC **210-0**, and the DBC **220-n** can be connected to the VDC **210-n**.

(32) In some embodiments, the DBC **220-0** can be configured to receive the digital signal VDC_0, and the DBC **220-n** can be configured to receive the digital signal VDC_n. In some embodiments, the DBC can convert the digital signal generated by the corresponding VDC to a binary signal. The DBC **220-0** can be configured to convert the digital signal VDC_0 to a binary signal DBC_0. The

DBC **220-n** can be configured to convert the digital signal VDC_n to a binary signal DBC_n. In some embodiments, the binary signals DBC₀ and DBC_n can be 1-bit, 2-bit, 3-bit, 4-bit . . . or 16-bit signal. In some embodiments, the binary signal can be more than 16 bits. In some embodiments, the bit count of the binary signal can be less than the bit count of the corresponding digital signal. For example, the DBC **220-0** can convert a 15-bit digital signal VDC₀ to a 4-bit binary signal DBC₀.

(33) In some embodiments, when the digital signal VDC₀ is 15-bit, being presented as VDC[14:0]=15'b000000000000000, the DBC **220-0** can convert it to a 4-bit binary signal DBC[3:0]=4'b0000. In some embodiments, the DBC **220-0** can convert the 15-bit digital signal VDC₀, being presented as VDC[14:0]=15'b0000000000000011, to a 4-bit binary signal DBC[3:0]=4'b0010.

(34) The binary signals DBC₀ and DBC_n can be transmitted to the adder **230**. In some embodiments, the adder **230** can be connected to the DBCs **220-0** and **220-n**. The adder **230** can be configured to receive two or more binary signals DBC₀, DBC₁, DBC₂ . . . and DBC_n. The adder **230** can be configured to generate an output signal Out[k:0] based on the binary signals DBC₀, DBC₁, DBC₂ . . . and DBC_n. The adder **230** can be configured to combine the binary signals DBC₀, DBC₁, DBC₂ . . . and DBC_n into an output signal Out[k:0]. In some embodiments, the bit count of the output signal Out[k:0] can exceed the bit count of the binary signals DBC₀ and DBC_n. For example, if the 4-bit binary signals DBC₀ and DBC_n are received by the adder **230**, the adder **230** can be configured to combine them into a 7-bit output signal Out[k:0]. As the signals are converted to the binary form, it is suitable for the adder **230** to combine them.

(35) The output signal Out[k:0] can be monitored and recorded. In some embodiments, the output signal Out[k:0] can be transmitted to the overshoot detector **240**. The overshoot detector **240** can be connected to the adder **230**. In some embodiments, the overshoot detector **240** can be configured to receive the output signal Out[k:0]. The overshoot detector **240** can be configured to detect a maximum value Out-max of the output signal Out[k:0]. The overshoot detector **240** can store the maximum value Out-max of the output signal Out[k:0]. In some embodiments, the maximum value Out-max of the output signal Out[k:0] can be transmitted to the MUX **260**. In some embodiments, the overshoot detector **240** can be configured to receive a reset signal.

(36) In some embodiments, the output signal Out[k:0] can be transmitted to the undershoot detector **250**. The undershoot detector **250** can be connected to the adder **230**. In some embodiments, the undershoot detector **250** can be configured to receive the output signal Out[k:0]. The undershoot detector **250** can be configured to detect a minimum value Out-min of the output signal Out[k:0]. The overshoot detector **240** can store the minimum value Out-min of the output signal Out[k:0]. In some embodiments, the minimum value Out-min of the output signal Out[k:0] can be transmitted to the MUX **260**. In some embodiments, the undershoot detector **250** can be configured to receive a reset signal.

(37) FIG. 3A is a flowchart **30A** showing operation of a voltage monitor, in accordance with some embodiments of the present disclosure. The operations of the overshoot detector **240** of the voltage monitor **20** are shown in FIG. 3A.

(38) Referring to FIG. 3A, the overshoot detector **240** can be reset in response to the reset signal at the beginning. Before detecting the output signal Out[k:0] from the adder **230**, the overshoot detector **240** can be set as an initial signal having the least value, in response to the reset signal. For example, the overshoot detector **240** can store an initial signal Overshoot[k:0]=k'b000 . . . 000. The overshoot detector **240** can compare the output signal Out[k:0] and the current value stored in the overshoot detector **240**. When the value of the output signal Out[k:0] is greater than the stored value, the output signal Out[k:0] is stored in the overshoot detector **240**. On the other hand, if the value of the output signal Out[k:0] is less than the stored value, the output signal Out[k:0] is not stored in the overshoot detector **240**. After the comparison, the overshoot detector **240** returns to

check if the reset signal is received, and continue operations for detecting the next output signal Out[k:0].

(39) The operation of FIG. 3A can be performed repeatedly by the overshoot detector **240**, such that the overshoot detector **240** can store a maximum value of the output signal Out[k:0].

(40) FIG. 3B is a flowchart **30B** including operation of a voltage monitor, in accordance with some embodiments of the present disclosure. The operations of the undershoot detector **250** of the voltage monitor **20** are shown in FIG. 3B. Referring to FIG. 3B, the undershoot detector **250** can be reset in response to the reset signal at the beginning. Before detecting the output signal Out[k:0] from the adder **230**, the undershoot detector **250** can be set as an initial signal having the greatest value, in response to the reset signal. For example, the undershoot detector **250** can store an initial signal Undershoot[k:0]-k'b111 . . . 111. The undershoot detector **250** can compare the output signal Out[k:0] and the current value stored in the undershoot detector **250**. When the value of the output signal Out[k:0] is less than the stored value, the output signal Out[k:0] is stored in the undershoot detector **250**. On the other hand, if the value of the output signal Out[k:0] is greater than the stored value, the output signal Out[k:0] is not stored in the undershoot detector **250**, and the undershoot detector **250** will be ready to receive the next output signal Out[k:0]. After the comparison, the undershoot detector **250** will go back to check if the reset signal is received, and continue operations for detecting the next output signal Out[k:0].

(41) The operation of FIG. 3B can be performed repeatedly by the undershoot detector **250**, such that the undershoot detector **250** can store a minimum value of the output signal Out[k:0].

(42) In some embodiments, the overshoot detector **240** and the undershoot detector **250** can be arranged in parallel as shown in FIG. 2A. That is, the overshoot detector **240** and the undershoot detector **250** can receive the output signal Out[k:0] in parallel, and then process the comparison.

(43) The MUX **260** can be connected to the adder **230**. In some embodiments, the MUX **260** can be configured to receive the current value of the output signal Out[k:0] from the adder **230**. The MUX **260** can be connected to the overshoot detector **240**. In some embodiments, the MUX **260** can be configured to receive the maximum value Out-max of the output signal Out[k:0] from the overshoot detector **240**. The MUX **260** can be connected to the undershoot detector **250**. In some embodiments, the MUX **260** can be configured to receive the minimum value Out-min of the output signal Out[k:0] from the undershoot detector **250**.

(44) The MUX **260** can be configured to receive three different values of the output signal Out[k:0]. The MUX **260** can be configured to receive a select signal sel[1:0]. In response to the select signal sel[1:0], the MUX **260** can output a MUX output signal MUX[k:0], which is selected from the current value, the maximum value Out-max, and the minimum value Out-min of the output signal Out[k:0].

(45) In some embodiments, the correspondence between the select signal sel[1:0] received by the MUX **260** and the MUX output signal MUX[k:0] is shown in Table 1 as follows.

(46) TABLE-US-00001 TABLE 1 sel[1:0] MUX[k:0] 00 Out[k:0] 01 Overshoot[k:0] 10 Undershoot[k:0]

(47) FIG. 2B shows an architecture of a voltage monitor **21** for monitoring the power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure. The voltage monitor **21** is similar to the voltage monitor **20** in FIG. 2A, with the difference therebetween being arrangement of the overshoot detector **240** and the undershoot detector **250**. Referring to FIG. 2B, the undershoot detector **250** is connected to the overshoot detector **240** to receive the output signal Out[k:0] from the adder **230** through the overshoot detector **240**. In some embodiments, the overshoot detector **240** and the undershoot detector **250** can be arranged in sequence as shown in FIG. 2B. That is, the overshoot detector **240** and the undershoot detector **250** can receive the output signal Out[k:0] in sequence, and then process the comparison.

(48) FIG. 3C is a flowchart **30C** showing operation of a voltage monitor, in accordance with some embodiments of the present disclosure. The operations of the overshoot detector **240** and the

undershoot detector **250** of the voltage monitor **21** are shown in FIG. 3C.

(49) Referring to FIG. 3C, the overshoot detector **240** and the undershoot detector **250** can be reset in response to the reset signal at the beginning. Before detecting the output signal Out[k:0], the overshoot detector **240** can be set as an initial signal having the least value, in response to the reset signal. For example, the overshoot detector **240** can store an initial signal Overshoot[k:0]=k'b000 . . . 000. The overshoot detector **240** can compare the output signal Out[k:0] and the current value stored in the overshoot detector **240**. The undershoot detector **250** can be set as an initial signal having the greatest value, in response to the reset signal. For example, the undershoot detector **250** can store an initial signal Undershoot[k:0]=k'b111 . . . 111.

(50) The overshoot detector **240** can receive the output signal Out[k:0] from the adder **230**. The overshoot detector **240** can compare the output signal Out[k:0] and the current value of the signal Overshoot[k:0] stored in the overshoot detector **240**. When the value of the output signal Out[k:0] is greater than the stored value, the output signal Out[k:0] is stored in the overshoot detector **240**. On the other hand, if the value of the output signal Out[k:0] is less than the stored value, the output signal Out[k:0] is not stored in the overshoot detector **240**. After the comparison, the overshoot detector **240** will transmit the output signal Out[k:0] to the undershoot detector **250**.

(51) The undershoot detector **250** can receive the output signal Out[k:0] from the overshoot detector **240**. The undershoot detector **250** can compare the output signal Out[k:0] and the current value of the signal Undershoot[k:0] stored in the undershoot detector **250**. When the value of the output signal Out[k:0] is less than the stored value, the output signal Out[k:0] is stored in the undershoot detector **250**. On the other hand, if the value of the output signal Out[k:0] is greater than the stored value, the output signal Out[k:0] is not stored in the undershoot detector **250**, and the undershoot detector **250** will be ready to receive the next output signal Out[k:0]. After the comparison, the undershoot detector **250** will go back to check if the reset signal is received, and continue operations for detecting the next output signal Out[k:0].

(52) FIG. 4A shows an architecture of a voltage-to-digital converter **210** included in the voltage monitor, in accordance with some embodiments of the present disclosure. In some embodiments, the voltage-to-digital converter (VDC) **210** can be included in the voltage monitor **20** or **21**. The VDC **210** in FIG. 4A can be any one of the VDCs **210-0** . . . and **210-n** in FIGS. 2A and 2B.

(53) The VDC **210** includes a controller **211** and one or more delay cells **215-0**, **215-1**, **215-2** . . . and **215-n**. In some embodiments, the delay cell **215-1** can include a multiplexer (MUX) **216**, a NAND gate **217**, and an inverter **218**. In some embodiments, the delay cells **215-1**, **215-2** . . . and **215-n** can have an arrangement similar to the delay cell **215-1**.

(54) Referring to FIG. 4A, the controller **211** is configured to receive the clock signal CLK. The controller **211** can be configured to generate an enable signal, a state signal, and a reset signal. In some embodiments, the controller **211** can be configured to generate the enable signal, the state signal, and the reset signal based on the clock signal CLK.

(55) The delay cell **215-0** is connected to the controller **211**. In some embodiments, the delay cell **215-0** can include a reset terminal configured to receive a reset signal, a first terminal configured to receive the enable signal, a second terminal configured to receive the state signal, and an output terminal configured to output an output signal VDC[0]. The delay cell **215-0** can be reset in response to the reset signal. In some embodiments, the delay cell **215-0** can generate the output signal VDC[0] in response to the enable signal and the state signal.

(56) The delay cell **215-1** is connected to the controller and the delay cell **215-0**. In some embodiments, the delay cell **215-1** can include a reset terminal configured to receive a reset signal, a first terminal configured to receive the output signal VDC[0] from the output terminal of the delay cell **215-0**, a second terminal configured to receive the state signal, and an output terminal configured to output an output signal VDC[1]. The delay cell **215-1** can be reset in response to the reset signal. In some embodiments, the delay cell **215-1** can generate the output signal VDC[1] in response to the enable signal and the state signal. In some embodiments, the delay cell **215-1** can

generate the output signal VDC[1] in response to the output signal VDC[0] and the state signal.

(57) The delay cell **215-2** is connected to the controller and the delay cell **215-1**. In some embodiments, the delay cell **215-2** can include a reset terminal configured to receive a reset signal, a first terminal configured to receive the output signal VDC[1] from the output terminal of the delay cell **215-1**, a second terminal configured to receive the state signal, and an output terminal configured to output an output signal VDC[2]. The delay cell **215-2** can be reset in response to the reset signal. In some embodiments, the delay cell **215-2** can generate the output signal VDC[2] in response to the enable signal and the state signal. In some embodiments, the delay cell **215-2** can generate the output signal VDC[2] in response to the output signal VDC[1] and the state signal.

(58) The delay cell **215-n** is connected to the controller and the previous delay cell **215-n-1** (not shown in FIG. 4A). In some embodiments, the delay cell **215-n** can include a reset terminal configured to receive a reset signal, a first terminal configured to receive the output signal VDC[n-1] (not shown in FIG. 4A) from the output terminal of the delay cell **215-n-1**, a second terminal configured to receive the state signal, and an output terminal configured to output an output signal VDC[n]. The delay cell **215-n** can be reset in response to the reset signal. In some embodiments, the delay cell **215-n** can generate the output signal VDC[n] in response to the enable signal and the state signal. In some embodiments, the delay cell **215-n** can generate the output signal VDC[n] in response to the output signal VDC[n-1] and the state signal.

(59) The output signal VDC[0] of the delay cell **215-0**, the output signal VDC[1] of the delay cell **215-1**, the output signal VDC[2] of the delay cell **215-2**, . . . and VDC[n] of the delay cell **215-n** are combined to be the output signal of the VDC **210**. Accordingly, the output signal of the VDC **210** can be an n-bit signal. The digital signal VDC_0 generated by the VDC **210-0** can be based on the output signal VDC[0] generated by the delay cell **215-0**, the output signal VDC[1] generated by the delay cell **215-1**, the output signal VDC[2] generated by the delay cell **215-2**, . . . and the output signal VDC[n] generated by the delay cell **215-n**. In some embodiments, the output signal of the VDC **210** can be VDC_0 . . . or VDC_n in FIG. 2A.

(60) Referring to FIG. 4A, the delay cell **215-0** includes the MUX **216**. The MUX **216** can be a two-to-one multiplexer. The MUX **216** can be connected to the controller **211**. In some embodiments, the MUX **216** can include a first input terminal, a second input terminal, a selector terminal, and an output terminal. The MUX **216** can be configured to receive the enable signal from the controller **211** through the first input terminal. The MUX **216** can be configured to receive the state signal from the controller **211** through the selector terminal.

(61) The NAND gate **217** of the delay cell **215-0** can be connected to the MUX **216** and the controller **211**. In some embodiments, the NAND gate **217** can include a first input terminal connected to the output terminal of the MUX **216**, a second input terminal, and an output terminal. The NAND gate **217** can be configured to receive the reset signal from the controller **211** through the second input terminal. In some embodiments, the NAND gate **217** can be any types of logic gates. For example, the logic gate can be NOR, OR, AND, etc.

(62) The inverter **218** of the delay cell **215-0** can be connected to the NAND gate **217**. The inverter **218** can be connected to the MUX **216**. The inverter **218** can include an input terminal connected to the output terminal of the NAND gate **217** and an output terminal connected to the second input terminal of the MUX **216**. The output terminal of the inverter **218** can be connected to the first terminal of the delay cell **215-2**. In some embodiments, the output signal VDC[0] can be obtained at the output terminal of the inverter **218**.

(63) According to the operations of the MUX **216**, the NAND gate **217** and the inverter **218** included in the delay cell **215-0**, the delay cell **215-0** can pass through or save the data based on the enable signal, state signal, and reset signal.

(64) FIG. 4B is a timing diagram of the voltage-to-digital converter **210** depicted in FIG. 4A, in accordance with some embodiments of the present disclosure. Referring to FIG. 4B, the clock signal CLK can be switched to logic high at the timing T1, and be switched to logic low at the

timing T3. The clock signal CLK can then be switched to logic high after being logic low for a constant period, such as at the timing T4 and T5.

(65) The controller **211** can be configured to generate the enable signal, the state signal, and the reset signal based on the clock signal CLK. The state signal can be substantially synchronized with the clock signal CLK. In some embodiments, the enable signal can be shifted a period (such as period P1) from the clock signal CLK. For example, the enable signal can be switched to logic high after the state signal switched to logic high for the period P1. In some embodiments, the enable signal can be switched to logic high at the timing T2. In some embodiments, the enable signal can have a frequency substantially identical to that of the clock signal CLK.

(66) In some embodiments, the reset signal can have a frequency substantially identical to that of the clock signal CLK. The reset signal can be switched to logic low at the timing T1, and be switched to logic high at the timing T2 after being logic low for the period P1.

(67) Referring back to FIG. 4A, the delay cell **215-0** of the VDC **210** can be configured to be reset during the period P1 in response to the reset signal. The delay cell **215-0** of the VDC **210** can be configured to be reset at the timing T1 in response to the reset signal. In some embodiments, the state signal and the reset signal can be switched at the timing T1. The delay cell **215-0** can be reset during the period P1 in response to the reset signal and the state signal. When all delay cells **215-0**, **215-1**, **215-2**, . . . , and **215-n** of the VDC **210** are reset, the output signal of the VDC **210** will be reset to 0.

(68) In some embodiment, the delay cell **215-0** can be configured to be triggered during a second period P2 in response to the enable signal. The delay cell **215-0** can be configured to be triggered at the timing T2. In some embodiments, the enable signal and the reset signal are switched to logic high at the same time.

(69) In some embodiments, the delay cells **215-0**, **215-1**, **215-2**, . . . , and **215-n** of the VDC **210** can be configured to be triggered during the period P2. FIG. 5A depicts an operating exemplary voltage-to-digital converter (VDC) **510A** included in the voltage monitor, in accordance with some embodiments of the present disclosure. The VDC **510A** is similar to the VDC **210** in FIG. 4A, with the difference therebetween being the lack of controller **211** (being omitted for clarity). In some embodiments, the VDC **510A** shows the operations of the delay cells during the period P2.

(70) Referring to FIG. 5A, the output signal VDC[0] of the delay cell **215-0** is in response to the enable signal. The output signal VDC[0] of the delay cell **215-0** can be transmitted to the delay cell **215-1**. In response to the output signal VDC[0], the delay cell **215-1** can output the output signal VDC[1]. The output signal VDC[1] of the delay cell **215-1** can be transmitted to the next delay cell. In some embodiments, the delay cell **215-n** can receive the output signal VDC[n-1] of the last delay cell. The delay cell **215-n** can output the output signal VDC[n] in response to the output signal VDC[n-1]. In some embodiments, each delay cell can have the same delay time.

(71) Referring back to FIG. 4B, the state signal can be switched to logic low at the timing T3. In some embodiments, the delay cell **215-0** can be configured to switch to a save state during a period P3 in response to the state signal. In some embodiments, the signals (data) can be saved in the delay cells at the timing T3.

(72) In some embodiments, the delay cells **215-0**, **215-1**, **215-2**, . . . , and **215-n** of the VDC **210** can be configured to switch to a save state during the period P3. FIG. 5B depicts an operating exemplary voltage-to-digital converter (VDC) **510B** included in the voltage monitor, in accordance with some embodiments of the present disclosure. The VDC **510B** is similar to the VDC **510A** in FIG. 5A, with the difference therebetween being the VDC **510B** showing the operations of the delay cells during the period P3.

(73) Referring to FIG. 5B, the output signal VDC[0] is saved in the delay cell **215-0**. The output signal VDC[1] is saved in the delay cell **215-1**. The output signal VDC[1] is saved in the delay cell **215-1**. The output signal VDC[n] is saved in the delay cell **215-n**. During the period P3, the output signal of the delay cell **215-0**, **215-1**, **215-2**, . . . , and **215-n** is not transmitted to the next delay cell.

In some embodiments, the output signal of the delay cell **215-0**, **215-1**, **215-2**, . . . , and **215-n** can be obtained (read) to be the digital signal (such as VDC_0 or VDC_n). The digital signal (for example, VDC_0) can be transmitted to the digital-to-binary converter (DBC) of the voltage monitor (for example, the DBC **220-0** in FIG. 2A).

(74) Referring back to FIG. 4B, the state signal can be switched to logic high at the timing T4. In some embodiments, the delay cells of the VBC can repeat the operations shown in FIGS. 5A and 5B according to each pulse of the clock signal CLK (or state signal). Referring to FIG. 2A, the digital signal VDC_0 of the VDC **210-0** saved in the period P3 can be transmitted to the DBC **220-0**, such that the digital signal VDC_0 can be converted to the binary signal DBC_0 at the timing T4.

(75) Referring back to FIG. 4B, the state signal can be switched to logic high at the timing T5 for the next pulse. Meanwhile, the delay cells of the VBC still repeat the operations shown in FIGS. 5A and 5B according to each pulse of the clock signal CLK (or state signal). Referring to FIG. 2A, the adder **230** can be configured to combine the binary signals DBC_0, DBC_1, . . . , DBC_n at the timing T5.

(76) In current practice, to avoid functional failure due to VDD droop, the supply voltage is increased. If the VDD droop can be detected accurately, the supply voltage is not overly increased, and thus can be reduced for power efficiency. The present disclosure provides a digital voltage monitor to make it easier to be integrated with the digital circuit. In other words, the provided voltage monitor is an all-digital circuit design optimizing integration with digital circuits. In addition, the voltage monitor **20** can achieve high resolution by adjusting the number of delay cells in the VDC according to need. The voltage monitor **20** depends on the clock signal CLK, such that the voltage detected by the voltage monitor **20** can have a high conversion speed aligned with clock speed. Due to the synchronization between the detected voltage and the clock signal, the current voltage can be monitored accurately. Moreover, the VDD detected by the voltage monitor **20** can be directly output in binary code, such that it will be easier to store and compare.

(77) FIG. 6 is a graph **60** illustrating time versus power voltage of a semiconductor device, in accordance with some embodiments of the present disclosure. Referring to FIG. 6, the x-axis represents the time in nanoseconds (ns). The y-axis represents the power voltage of the semiconductor device in volts (V).

(78) The solid line, marked voltage of Out[k:0], shows the voltage recorded by the voltage monitor (such as the voltage monitor **20** in FIG. 2A). Since the voltage monitor **20** saves the signals/data in binary code, they can be transformed back to voltage value. Therefore, the voltage values recorded by the voltage monitor **20** are shown in the graph **60**. The dash line, marked VDD, may show the simulated power voltage (VDD) of the semiconductor device, which includes the voltage monitor **20**.

(79) As shown in FIG. 6, the first trough, which indicates the first VDD droop, occurs at about 1 ns. The second trough, indicating the second VDD droop, occurs at about 6 ns. The third trough, indicating the third VDD droop, occurs at about 10.5 ns. Overall, the trade of the solid line substantially conforms to that of the dash line. That is, the voltage monitor **20** is accurate and thus can be applicable to monitor the VDD droop.

(80) FIG. 7 is a graph **70** illustrating power voltage (VDD) versus binary signal of the recorded voltage of a semiconductor device, in accordance with some embodiments of the present disclosure. Referring to FIG. 7, the x-axis represents the power voltage (VDD) of the semiconductor device in voltage (V). The y-axis represents the binary signal Out[k:0] of the recorded voltage of the semiconductor device. The graph **70** includes three lines **701**, **702**, and **703**.

(81) The line **701** shows the binary signal Out[k:0] of the recorded voltage of the voltage monitor located in a region (corner) of the semiconductor device operating at high frequency under low temperatures. In some embodiments, the line **701** indicates the voltage monitor located in the region fast operating under a temperature of -25°C . The line **702** shows the binary signal Out[k:0]

of the recorded voltage of the voltage monitor located in a region (corner) of the semiconductor device operating at low frequency under high temperatures. In some embodiments, the line **702** indicates the voltage monitor located in the region slow operating under a temperature of 125° C. The line **703** shows the binary signal Out[k:0] of the recorded voltage of the voltage monitor located in a region (corner) of the semiconductor device operating at low frequency under a low temperature. In some embodiments, the line **703** indicates the voltage monitor located in the region slow operating under a temperature of -25° C.

(82) The lines **701**, **702**, and **703** can be obtained from a simulation. In some embodiments, the lines **701**, **702**, and **703** can be obtained by statistics. The lines **701**, **702**, and **703** show that different regions of the same semiconductor device can have different characteristics. For example, the lines **701**, **702**, and **703** show that different regions of the same semiconductor device can have different correspondence between the input power voltage and the binary output signal Out[k:0]. Therefore, one or more voltage monitors can be disposed in one semiconductor device, such that the VDD droops of the different regions of the semiconductor device, which may differ from each other, can be monitored and recorded.

(83) FIG. **8A** shows an exemplary method **80A** for calibrating the voltage monitor **20** of the semiconductor device, in accordance with some embodiments of the present disclosure. The method **80A** utilizes the graph **70** shown in FIG. **7** to calibrate the voltage monitors located in the semiconductor device.

(84) The method **80A** can be 1-point calibration. Referring to FIG. **8A**, the method **80A** includes Step 1 and Step 2. In Step 1, the VDD is set to a particular value (for example, 0.8V), and the corresponding binary signal Out[k:0] (for example, 60) can be obtained. In Step 2, the corresponding line in the graph **70** (from the lines **701**, **702**, and **703**) is determined according to the correspondence between the input VDD and the obtained output binary signal Out[k:0]. In some embodiments, as the corresponding line in the graph **70** can be determined, the location of the voltage monitor in the semiconductor device can be determined accordingly. Once the slope of the corresponding line is determined, other binary signals Out[k:0] of the same voltage monitor can be predicted according to the slope of the corresponding line in the graph **70**. In some embodiments, if the region to be calibrated is operated at high frequency under -25° C., the slope of the line **701** should be obtained so as to predict the corresponding binary signal Out[k:0].

(85) FIG. **8B** shows an exemplary method **80B** for calibrating the voltage monitor of the semiconductor device, in accordance with some embodiments of the present disclosure. The method **80B** utilizes the graph **70** shown in FIG. **7** to calibrate the voltage monitors located in the semiconductor device.

(86) The method **80B** can be 2-point calibration. Referring to FIG. **8B**, the method **80B** includes Steps 1, 2, 3, and 4. In Step 1, the VDD is set to a first value (for example, 0.8V), and the corresponding binary signal Out[k:0] (for example, 60) can be obtained. In Step 2, the VDD is set to a second value (for example, 0.7V), and the corresponding binary signal Out[k:0] (for example, 45) can be obtained. In Step 3, the slope of the corresponding line (the inclined line approaching the line **701**) is calculated according to the two points in Steps 1 and 2. In some embodiments, as the corresponding line can be determined, the location of the voltage monitor in the semiconductor device can be determined accordingly based on the graph **70**. In Step 4, other binary signals Out[k:0] can be calculated based on the calculated slope of the corresponding line, in response to the VDD. Since the slope of the corresponding line is determined, other binary signals Out[k:0] of the same voltage monitor can be predicted according to the slope of the corresponding line.

(87) FIG. **8C** shows an exemplary method **80C** for calibrating the voltage monitor of the semiconductor device, in accordance with some embodiments of the present disclosure.

(88) The method **80C** can be multi-point calibration. Referring to FIG. **8C**, the method **80C** includes Steps 1, 2, 3, and 4, and then repeats Steps 1 through 4 for other voltage ranges. In Step 1, the VDD is set to a first value (for example, 0.8V), and the corresponding binary signal Out[k:0]

(for example, 60) can be obtained. In Step 2, the VDD is set to a second value (for example, 0.7V), and the corresponding binary signal Out[k:0] (for example, 45) can be obtained. In Step 3, the slope of corresponding line (the inclined line approaching the line **701**) is calculated according to the two points in Steps 1 and 2. In some embodiments, as the corresponding line can be determined, the location of the voltage monitor in the semiconductor device can be determined accordingly based on the graph **70**. In Step 4, other binary signals Out[k:0], in response to the VDD only in the range of 0.7 to 0.8V, can be calculated based on the slope of the corresponding line.

(89) For the VDD out of the particular range (0.7 to 0.8V), the aforementioned Steps 1 to 4 should repeat to obtain a line corresponding to desired range. For example, if the binary signal Out[k:0] corresponding to the VDD of 0.65V is to be calculated, the Steps 1 to 4 are performed under the VDD in the range of 0.6 to 0.7V.

(90) After utilizing the methods shown in FIGS. **8A-8C** calibrating the region to be tested of the semiconductor device, the VDD droop can be detected and recorded by the voltage monitor. In one embodiment, it can be determined whether the monitored voltage droop exceeds a threshold so as to avoid function failure. If the VDD droop is determined to have exceeded a threshold, such as 10% VDD, the chips adjacent to the voltage monitor can be controlled to reduce activities and clock speed. In some embodiments, if the VDD droop is determined to exceed a threshold, such as 10% VDD, the voltage regulator, which provides the power voltage (VDD) to the semiconductor device, can be controlled to increase the VDD.

(91) In another embodiment, the monitored voltage can be utilized to predict the VDD droop so as to avoid function failure. If the monitored VDD slope is greater than a threshold, the chips adjacent to the voltage monitor can be controlled to reduce activities and clock speed in advance. If the monitored VDD slope (which may be calculated based on two recorded values) is greater than a threshold, the voltage regulator, which provides the power voltage (VDD) to the semiconductor device, can be controlled to increase the VDD in advance.

(92) FIG. **9** is a top view of an exemplary semiconductor device **90**, in accordance with some embodiments of the present disclosure. The semiconductor device **90** can include chips (or semiconductor dies), voltage monitors **920A**, **920B**, and **920C**, and power delivery network. In some embodiments, the semiconductor device **90** can include a substrate (not shown). The chips and the voltage monitors **920A**, **920B**, and **920C** can be disposed on the substrate. One or more voltage monitors **920A**, **920B**, and **920C** can be disposed within the chips. In some embodiments, the voltage monitors **920A**, **920B**, and **920C** can be located in the regions of the semiconductor device **90**, which may operate in different frequency. The power delivery network can connect the chips to deliver power. In some embodiments, the power delivery network can connect the chips and the voltage monitors to provide power supply voltage to them.

(93) In some embodiments, the voltage monitors **920A**, **920B**, and **920C** can determine the power supply voltage of the chips adjacent thereto. As shown in FIG. **9**, the hot spot **950** may occur about the voltage monitor **920C**. In some embodiments, the hot spot **950** may be caused by the large VDD droop. Since the chips may functionally fail due to large VDD droop, the hot spot **950** may therefore occur. As the voltage monitor **920C** is disposed near the hot spot **950**, the function failure can be avoided by monitoring the VDD droop.

(94) In one embodiment, the present disclosure provides a voltage monitor. The voltage monitor includes a first voltage-to-digital converter (VDC), a second VDC, a first digital-to-binary converter (DBC), a second DBC, and an adder. The first VDC is configured to generate a first digital signal in response to a clock signal, and the second VDC is configured to generate a second digital signal in response to the clock signal. The first DBC is connected to the first VDC, and configured to convert the first digital signal to a first binary signal. The second DBC is connected to the second VDC, and configured to convert the second digital signal to a second binary signal. The adder is connected to the first DBC and the second DBC, and configured to combine the first binary signal and the second binary signal into an output signal.

(95) In another embodiment, the present disclosure provides a voltage monitor. The voltage monitor includes a plurality of voltage-to-digital converters (VDC), a plurality of digital-to-binary converters (DBC), and an adder connected to the plurality of DBCs. Each of the plurality of VDCs is configured to generate a respective digital signal in response to a clock signal. Each of the plurality of DBCs is connected to a corresponding VDC of the plurality of VDCs, and configured to convert each of the respective digital signals to a respective binary signal. The adder is configured to combine the respective binary signals of the plurality of the DBCs into an output signal.

(96) In some embodiments, the present disclosure provides a semiconductor device. The semiconductor includes a substrate, a plurality of semiconductor dies disposed on the substrate, and a voltage monitor disposed on the substrate, such that the voltage monitor determines a power supply voltage of the semiconductor dies adjacent to the voltage monitor.

(97) The methods and features of the present disclosure have been sufficiently described in the above examples and descriptions. It should be understood that any modifications or changes without departing from the spirit of the present disclosure are intended to be covered in the protection scope of the present disclosure.

(98) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As those skilled in the art will readily appreciate from the present disclosure, processes, machines, manufacture, composition of matter, means, methods or steps presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein, can be utilized according to the present disclosure.

(99) Accordingly, the appended claims are intended to include within their scope: processes, machines, manufacture, compositions of matter, means, methods or steps. In addition, each claim constitutes a separate embodiment, and the combination of various claims and embodiments are within the scope of the present disclosure.

Claims

1. A voltage monitor, comprising: a first voltage-to-digital converter (VDC) configured to generate a first digital signal in response to a clock signal; a second VDC configured to generate a second digital signal in response to the clock signal; a first digital-to-binary converter (DBC) connected to the first VDC, and configured to convert the first digital signal to a first binary signal; a second DBC connected to the second VDC, and configured to convert the second digital signal to a second binary signal; and an adder connected to the first DBC and the second DBC, and configured to combine the first binary signal and the second binary signal into an output signal.
2. The voltage monitor of claim 1, further comprising: an overshoot detector connected to the adder, and configured to detect and store a maximum value of the output signal.
3. The voltage monitor of claim 2, further comprising: an undershoot detector connected to the adder, and configured to detect and store a minimum value of the output signal.
4. The voltage monitor of claim 3, further comprising: a multiplexer (MUX) connected to the adder, the overshoot detector, and the undershoot detector, wherein the MUX is configured to output one of: a current value of the output signal from the adder, the maximum value of the output signal from the overshoot detector, or the minimum value of the output signal from the undershoot detector.
5. The voltage monitor of claim 1, wherein the first VDC comprises: a controller configured to receive the clock signal, and generate an enable signal, a state signal, and a reset signal; a first delay cell connected to the controller, wherein the first delay cell has a reset terminal configured to receive a reset signal, a first terminal configured to receive the enable signal, a second terminal

configured to receive the state signal, and an output terminal configured to output a first output signal; and a second delay cell connected to the controller, the second delay cell has a reset terminal configured to receive a reset signal from the controller, a first terminal configured to receive the first output signal from the first delay cell, a second terminal configured to receive the state signal from the controller, and an output terminal configured to output a second output signal.

6. The voltage monitor of claim 5, wherein the first delay cell is configured to: be reset during a first period in response to the reset signal; be triggered during a second period in response to the enable signal; and switch to a save state during a third period in response to the clock signal.

7. The voltage monitor of claim 5, wherein the first digital signal generated by the first VDC is based on the first output signal generated by the first delay cell and the second output signal generated by the second delay cell.

8. The voltage monitor of claim 5, wherein the first delay cell comprises: a two-to-one multiplexer (MUX) including a first input terminal configured to receive the enable signal, a second input terminal, a selector terminal configured to receive the state signal, and an output terminal; a logic gate including a first input terminal connected to the output terminal of the MUX, a second input terminal configured to receive the reset signal, and an output terminal; and an inverter including an input terminal connected to the output terminal of the logic gate and an output terminal connected to the second input terminal of the MUX.

9. The voltage monitor of claim 1, wherein the first VDC includes a plurality of delay cells, wherein the second VDC includes a plurality of delay cells, and wherein the delay cells of the first VDC has a delay time different from that of the delay cells of the second VDC.

10. A semiconductor device comprising: a substrate; a plurality of semiconductor dies disposed on the substrate; and the voltage monitor according to claim 1 disposed on the substrate, such that the voltage monitor determines a power supply voltage of the semiconductor dies adjacent to the voltage monitor.

11. A voltage monitor, comprising: a plurality of voltage-to-digital converters (VDC), each configured to generate a respective digital signal in response to a clock signal; a plurality of digital-to-binary converters (DBC), each connected to a corresponding VDC of the plurality of VDCs, and configured to convert each of the respective digital signals to a respective binary signal; an adder connected to the plurality of DBCs, and configured to combine the respective binary signals of the plurality of the DBCs into an output signal.

12. The voltage monitor of claim 11, further comprising: an overshoot detector connected to the adder and configured to: (i) receive the output signal; (ii) compare the output signal and a first initial signal stored in the overshoot detector; and (iii) store the output signal in the overshoot detector when the output signal exceeds the first initial signal.

13. The voltage monitor of claim 12, wherein the overshoot detector is further configured to repeat the operations (i) to (iii) such that the overshoot detector store a maximum value of the output signal.

14. The voltage monitor of claim 13, further comprising: an undershoot detector connected to the adder, the undershoot detector configured to (a) receive the output signal; (b) compare the output signal and a second initial signal stored in the undershoot detector; and (c) store the output signal in the undershoot detector when the output signal is less than the second initial signal.

15. The voltage monitor of claim 14, wherein the undershoot detector is further configured to repeat the operations (a) to (c) such that the undershoot detector store a minimum value of the output signal.

16. The voltage monitor of claim 15, further comprising: a multiplexer (MUX) connected to the adder, the overshoot detector, and the undershoot detector, wherein the MUX is configured to output one of: a current value of the output signal from the adder, the maximum value of the output signal from the overshoot detector, or the minimum value of the output signal from the undershoot detector.

17. The voltage monitor of claim 11, wherein each of the plurality of VDCs comprises: a controller configured to receive the clock signal, and generate an enable signal, a state signal, and a reset signal; a first delay cell connected to the controller and configured to receive the enable signal, the state signal, and the reset signal, wherein the first delay cell is configured to generate a first output signal in response to the enable signal, the state signal, and the reset signal; and a second delay cell connected to the controller and the first delay cell, the second delay cell configured to receive the state signal and the reset signal from the controller and to receive the first output signal from the first delay cell, wherein the second delay cell is configured to generate a second output signal in response to the first output signal, the state signal, and the reset signal.

18. The voltage monitor of claim 17, wherein the first delay cell is configured to: be reset during a first period in response to the reset signal; be triggered during a second period in response to the enable signal; and switch to a save state during a third period in response to the clock signal.

19. The voltage monitor of claim 17, wherein the respective digital signal generated by each of the plurality of VDCs is based on the first output signal generated by the first delay cell and the second output signal generated by the second delay cell.

20. The voltage monitor of claim 17, wherein the first delay cell comprises: a two-to-one multiplexer (MUX) including a first input terminal configured to receive the enable signal, a second input terminal, a selector terminal configured to receive the state signal, and an output terminal; a logic gate including a first input terminal connected to the output terminal of the MUX, a second input terminal configured to receive the reset signal, and an output terminal; and an inverter including an input terminal connected to the output terminal of the logic gate and an output terminal connected to the second input terminal of the MUX.
